



Lecture example – Calculation of stresses in an uncracked prestressed beam

A prestressed concrete beam spans 10 m and has a rectangular cross-section 800 mm deep and 400 mm wide. It is post-tensioned by a cable having a cross-sectional area of 1000 mm². The cable eccentricity varies parabolically from zero at each end to a maximum of 250 mm at midspan, where the prestressing force is 1200 kN. $f'_c = 32$ MPa

Determine

1. The stresses at mid-span due to prestress acting alone
2. The decompression moment
3. The stress at midspan due to prestress, self weight and a 30 kN/M load

The cross section properties are:

Cross-sectional area $A_g = 320,000 \text{ mm}^2$

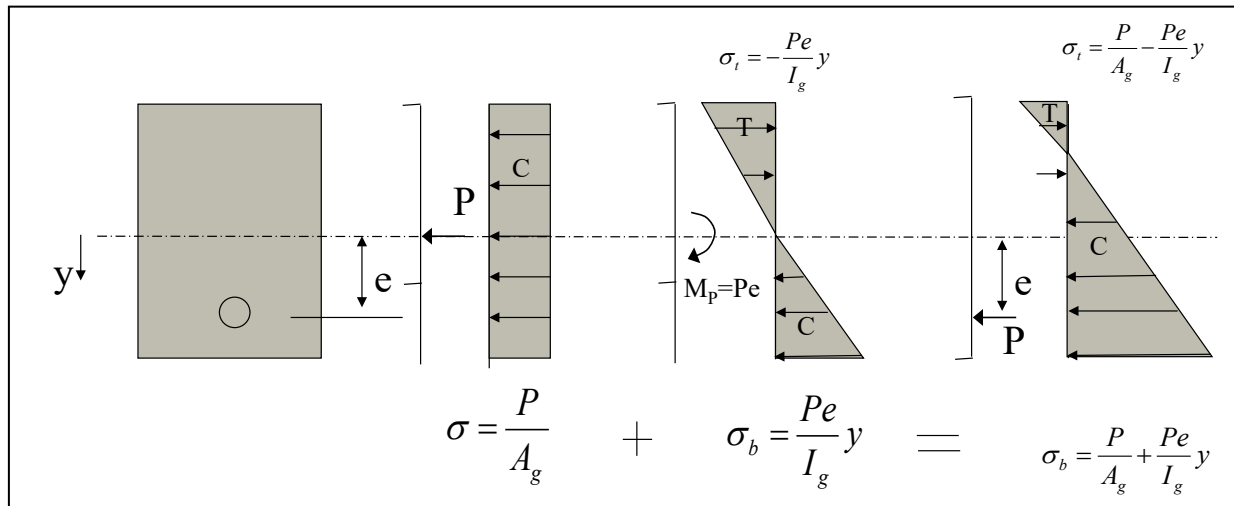
Second moment of area $I_g = 17.07 \times 10^9 \text{ mm}^4$;

$Z = bD^2/6 = 42.67 \text{ mm}^3$

For a concrete unit weight of 25 kN/m³, the self -weight is $25 \times 0.32 = 8 \text{ kN/m}$

1) Stresses due to prestress alone

At the mid-span section where the cable eccentricity is 250mm, the concrete stress is equivalent to an axial compressive force $N = 1200\text{kn}$, and a couple $M = Pe = 1200 \times 10^3 \times 250 = 300 \times 10^6 \text{ Nmm}$



Stresses in concrete due to prestress

At the top of the fibre of the beam:

$$\sigma_t = \frac{P}{A_g} - \frac{P.e.y_t}{I_g} = \frac{1200 \times 10^3}{320,000} - \frac{1200 \times 10^3 \times 250 \times 400}{17.07 \cdot 10^9} = -3.28 \text{ MPa (tension)}$$

(note top of beam is in tension)

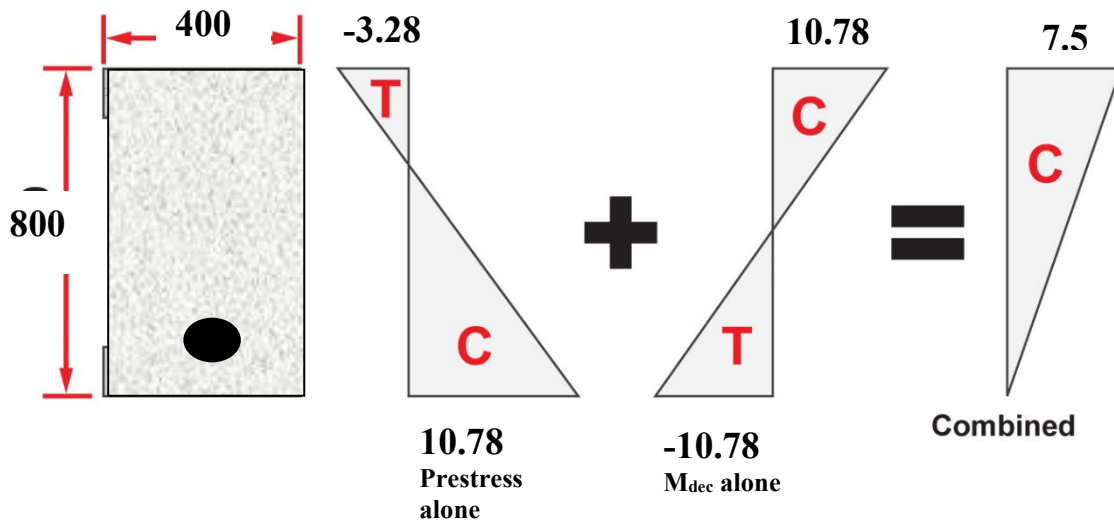
At the bottom of the fibre of the beam:

$$\sigma_b = \frac{P}{A_g} + \frac{P.e.y_b}{I_g} = \frac{1200 \times 10^3}{320,000} + \frac{1200 \times 10^3 \times 250 \times 400}{17.07 \cdot 10^9} = 10.78 \text{ MPa (compression)}$$

2) Decompression moment at midspan

To achieve decompression; i.e. zero bottom fibre stress, the bending moment M_{dec} must reduce the bottom fibre compressive stress of 10.78 MPa due to prestress, to zero. M_{dec} is therefore the bending moment which by itself would induce a bottom fibre tensile stress of 10.28 MPa.

$$M_{dec} = \frac{\sigma_{bp} \cdot I_g}{y_b} = \frac{10.78 \times 17.07 \times 10^9}{400} = 460 \text{ kNm}$$



Decompression moment at midspan,

3) Stresses at mid-span due to prestress, self-weight and live load

Self -weight = 8 kN/m

Live load = 30 kN/m

Total applied load = 8 + 30 = 38 kN/m

Total applied moment = $M_{app} = \frac{38 \times 10^2}{8} = 475 \text{ kNm}$

At top fibre

$$\sigma_t = \left(\frac{P}{A_g} - \frac{P \cdot e \cdot y_t}{I_g} \right) + \frac{M_{app} y_t}{I_g} = -3.28 + \frac{475 \times 10^6 \times 400}{17.07 \times 10^9} = 7.85 \text{ MPa (compression)}$$

At bottom fibre

$$\sigma_b = \left(\frac{P}{A_g} + \frac{P \cdot e \cdot y_b}{I_g} \right) - \frac{M_{app} y_b}{I_g} = 10.78 - \frac{475 \times 10^6 \times 400}{17.07 \times 10^9} = -0.35 \text{ MPa (tension)}$$

It can be seen that the full service load moment is only just greater than the decompression moment

Prestressed Flexural Members

Problem 1 – Elastic stress distribution

Calculate the elastic stress distribution for the beam shown below. (That is, calculate the extreme concrete fibre stresses at midspan due to prestressing and applied loads)

:

- The applied prestressing force on the beam, $P = 2400 \text{ kN}$
- The uniformly distributed load including self-weight = 30 kN / m
- The area of beam cross section, $A = 320,375 \text{ mm}^2$
- The second moment of inertia, $I_g = 5.02 \times 10^{10} \text{ mm}^4$

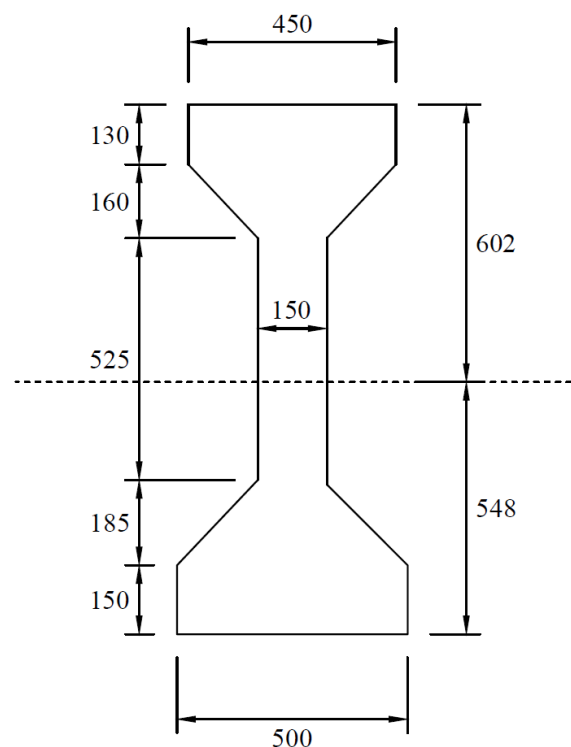


Figure 1.1 - Beam cross-section

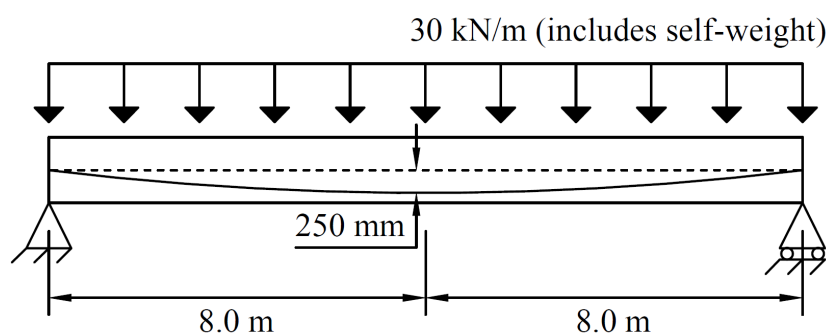


Figure 1.2 – Beam elevation

Problem 2 – Calculation of stresses in an uncracked prestressed beam

The cross-section of a post-tensioned concrete beam spanning 15 m, simply supported, is shown in Figure 2.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 300 mm. The tendon comprises of 8 number 12.7mm strands. The tendon is stressed to 80% of the breaking load. Properties of the materials are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

$$\text{specific weight of concrete} = 25 \text{ kN} / \text{m}^3$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

1. Determine the stresses in the beam section at midspan due to prestress acting alone
2. Determine the decompression moment at midspan
3. Determine the stresses at a midspan due to prestress, self-weight and a uniform live load of $18 \text{ kN} / \text{m}$.
4. Is the beam cracked under the above loading?

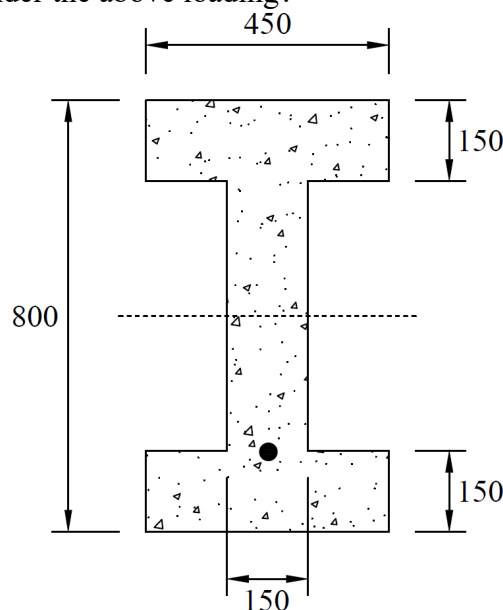


Figure 1.1 - Uncracked prestressed beam section

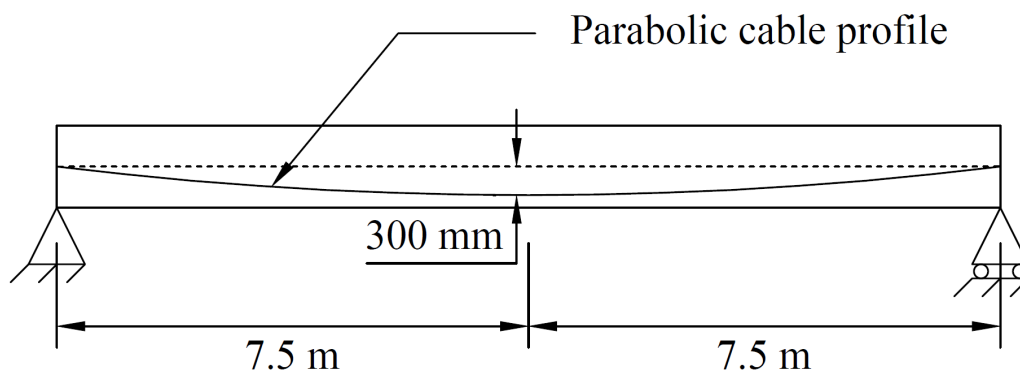


Figure 2.2 - Post-tensioning cable profile